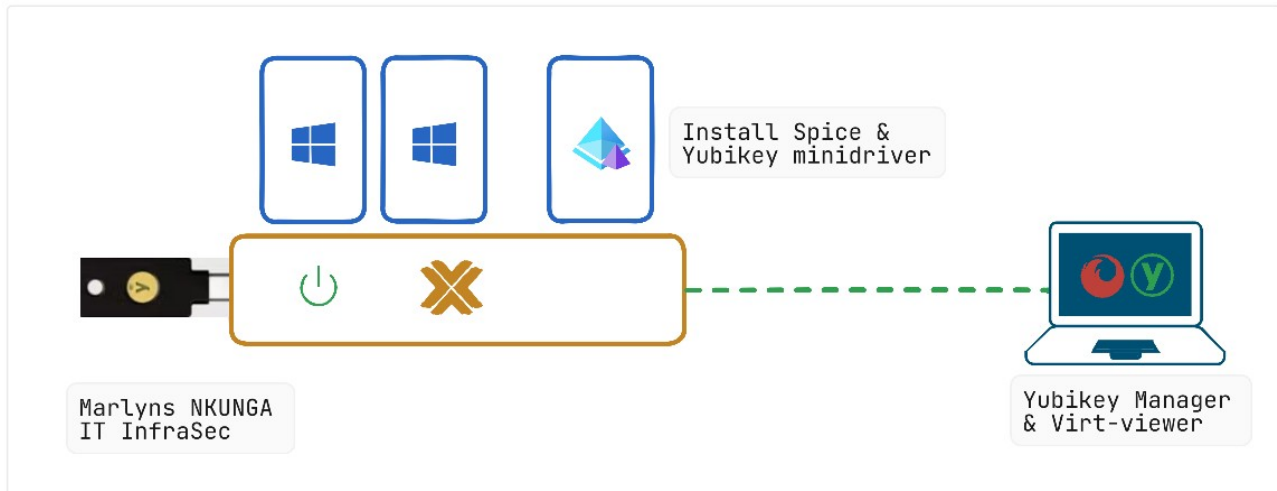


# **Moving beyond password to passwordless**

## Overview

The main goal is to configure Active Directory Passwordless Authentication with Yubikey, for the hardware constraint, I used Proxmox VE to complete this project, there are some requirements must be met on the client virtual machines. You'll find these tools in the tools folder.



**Notice that,** **Virt-viewer** and **Yubikey manager** must be installed on the management machine, Yubikey Manager to prepare the Yubikey, such as change the PIN, PUK and disable unused protocols.

### First step : Prepare your virtual machines,

- Install OS, IP addresses configuration
- Promote a domain controller

### Second step : requirements installation

These tools must be installed on your virtual machines, it's simple just next over and over.

- Spice Guest Tool
- Yubikey minidriver
- Yubikey Manager
- Virt-viewer
- Yubikey 5C NFC Serie

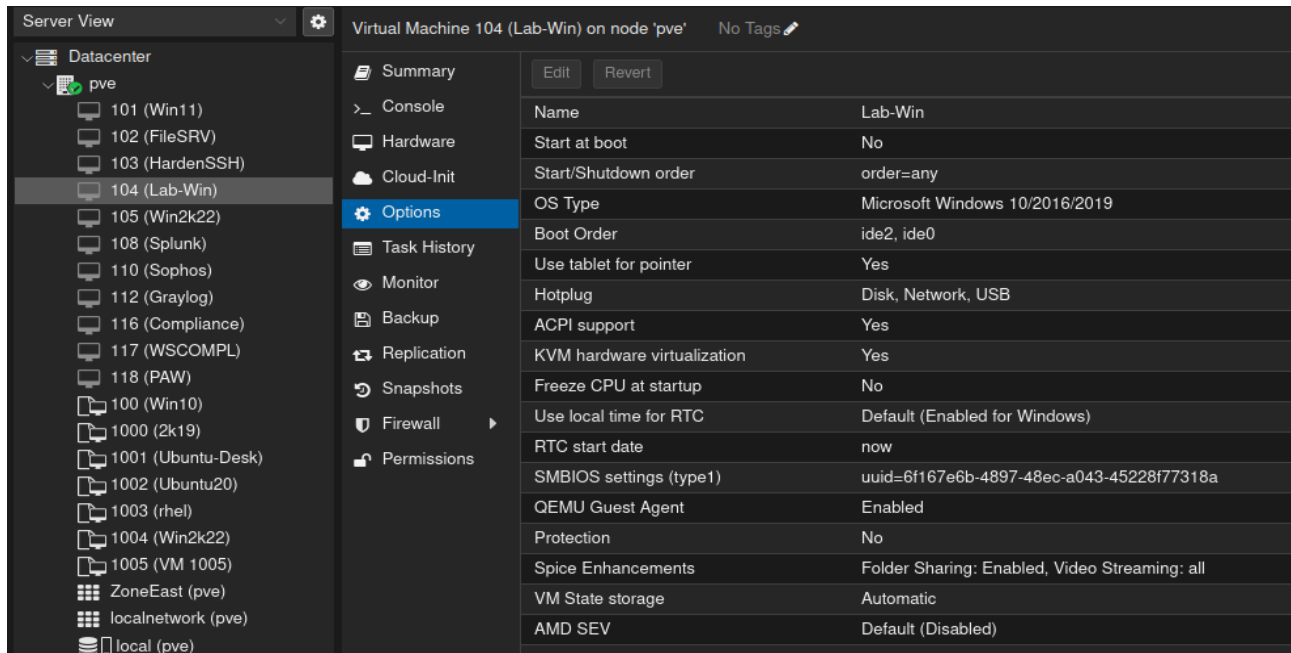
**Notice that,** If you'll use smartcard authentication via RDP, **Yubikey minidriver** must be installed with a specific parameter on your servers. You'll find a PowerShell in tools folder, it helps you to install **Yubikey minidriver** in legacy mode on the servers.

**msiexec /i YubiKey-Minidriver-latest-x64.msi INSTALL\_LEGACY\_NODE=1 /quiet**

### Third step : Spice Guest tool configuration on the virtual machines

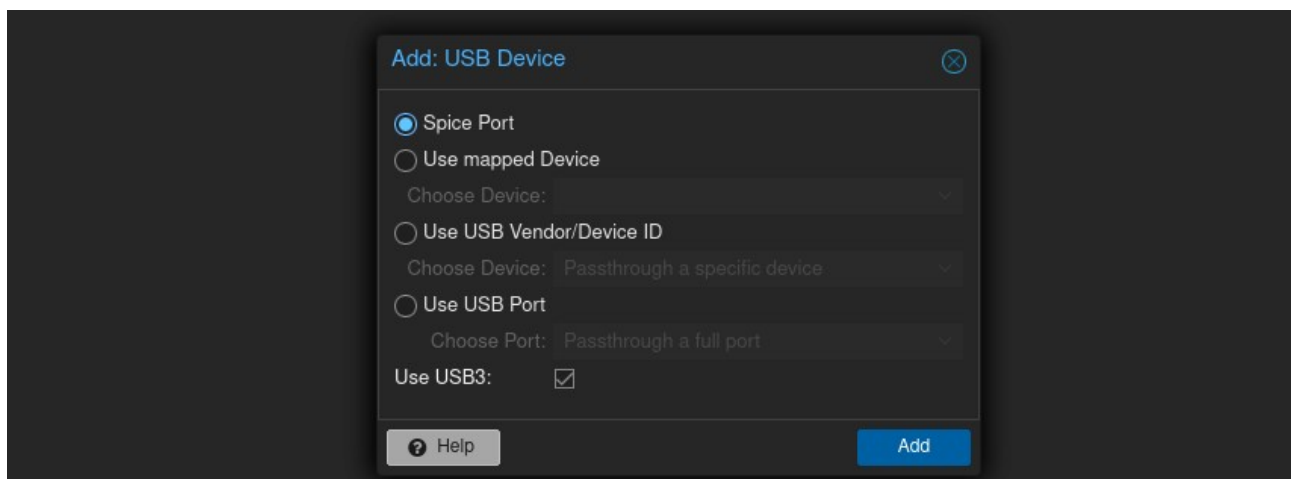
After configuring this tool the VM will be able to detect the Yubikey, when it'll be plugged.

Select the VM, Navigate to Options > Edit QEMU Guest Agent > select Use QEMU Guest Agent > Ok

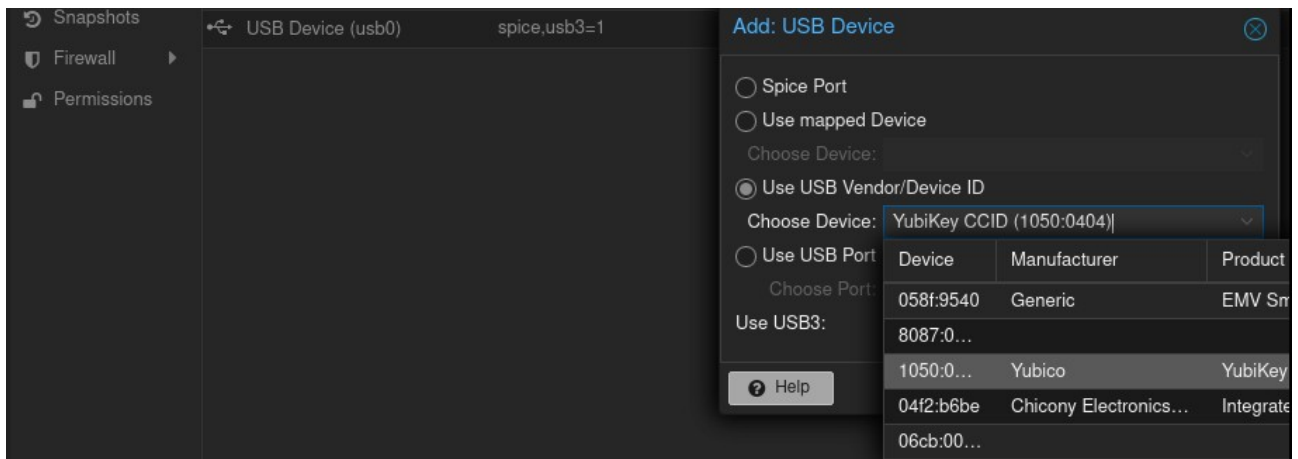


Notice that : QEMU Guest Agent must be enabled, as you see on the screenshot

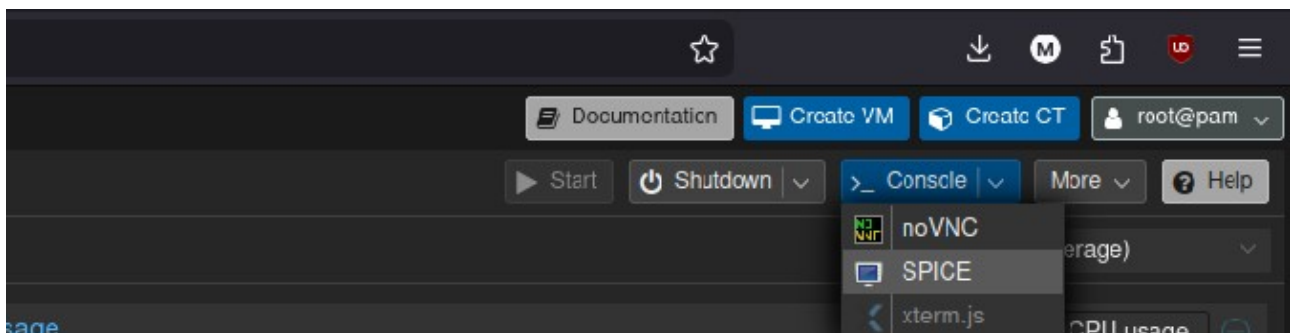
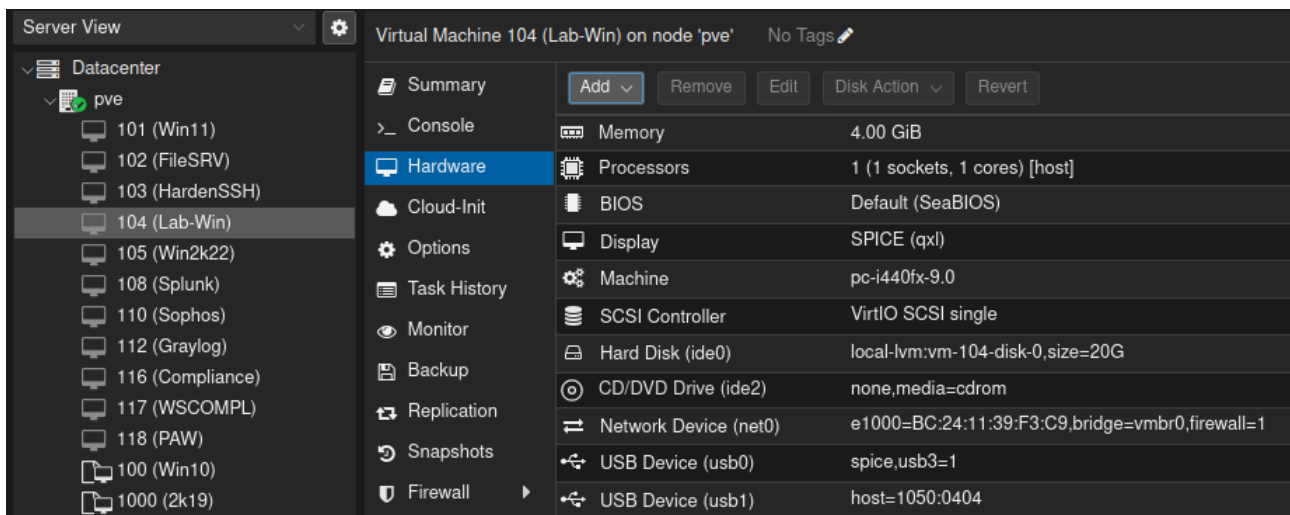
Select the VM > Navigate to Hardware > Add > USB Device > Select Spice Port and Use USB3 > Add



Select the VM > Navigate to Hardware > Add > USB Device > Select Use USB Vendor/Device ID > Yubico > Add



## Final result

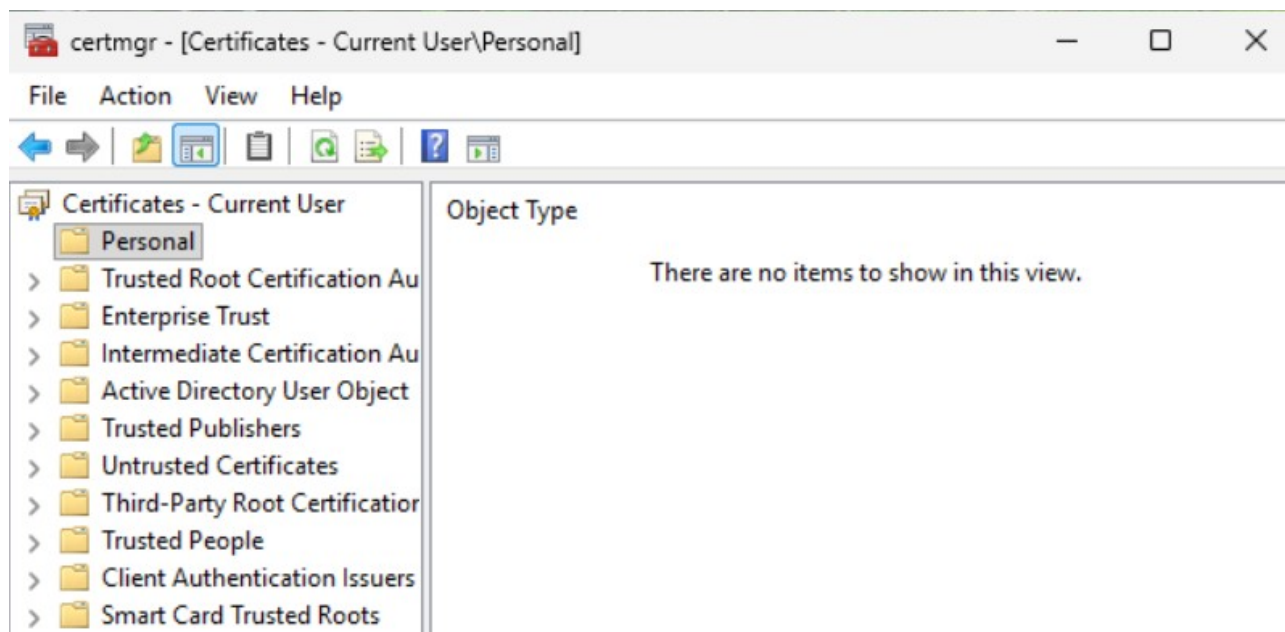


**Notice that,** Spice need the virt-viewer application on the management machine

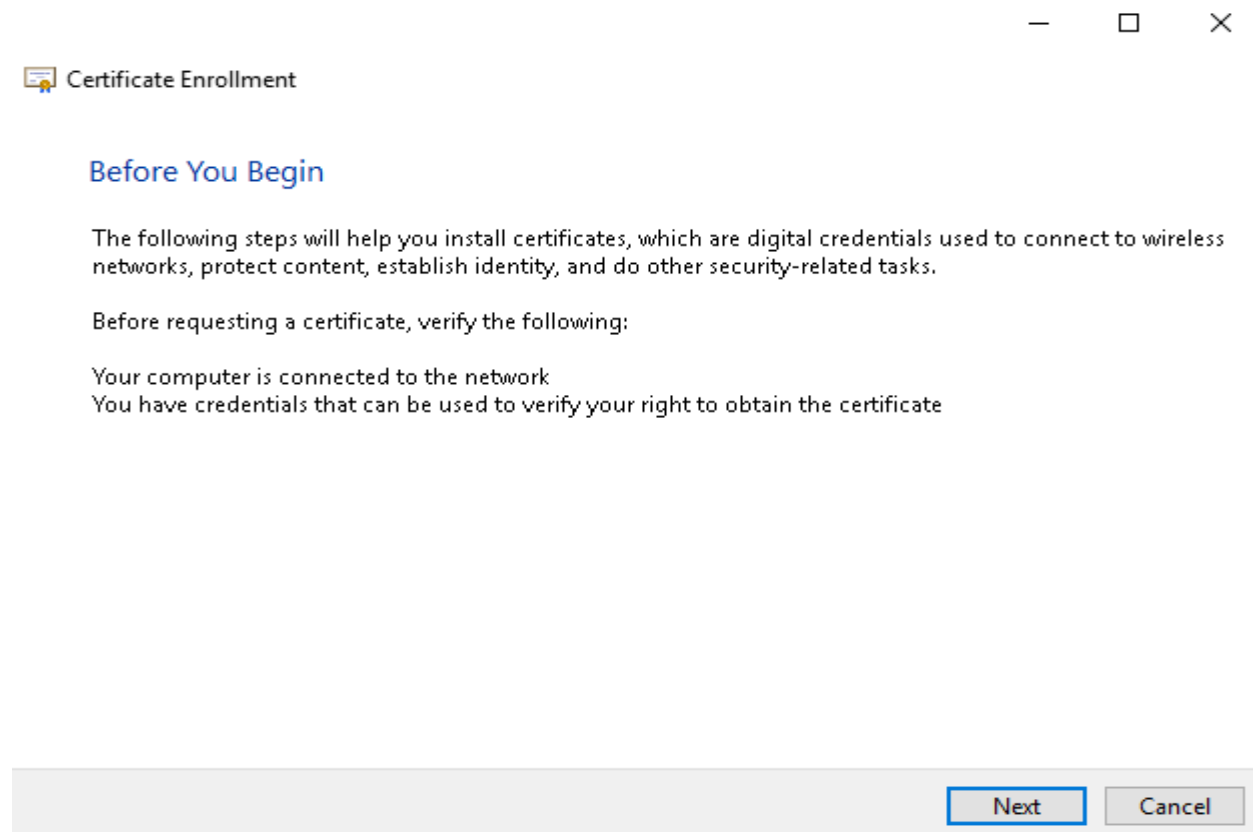
### Fourth Step : Enrollment

Quickadcs script will create a security group called **Yubikey Authentication**, only the member of this group can enroll to Yubikey Smartcard logon certificate

Launch this command **certmgr.msc** in Execute



Navigate to Certificates > Personal Right Click > All task > New Request



 Certificate Enrollment

### Select Certificate Enrollment Policy

Certificate enrollment policy enables enrollment for certificates based on predefined certificate templates. Certificate enrollment policy may already be configured for you.

**Configured by your administrator**  
Active Directory Enrollment Policy 

**Configured by you** [Add New](#)

Next

Cancel

 Certificate Enrollment

### Request Certificates

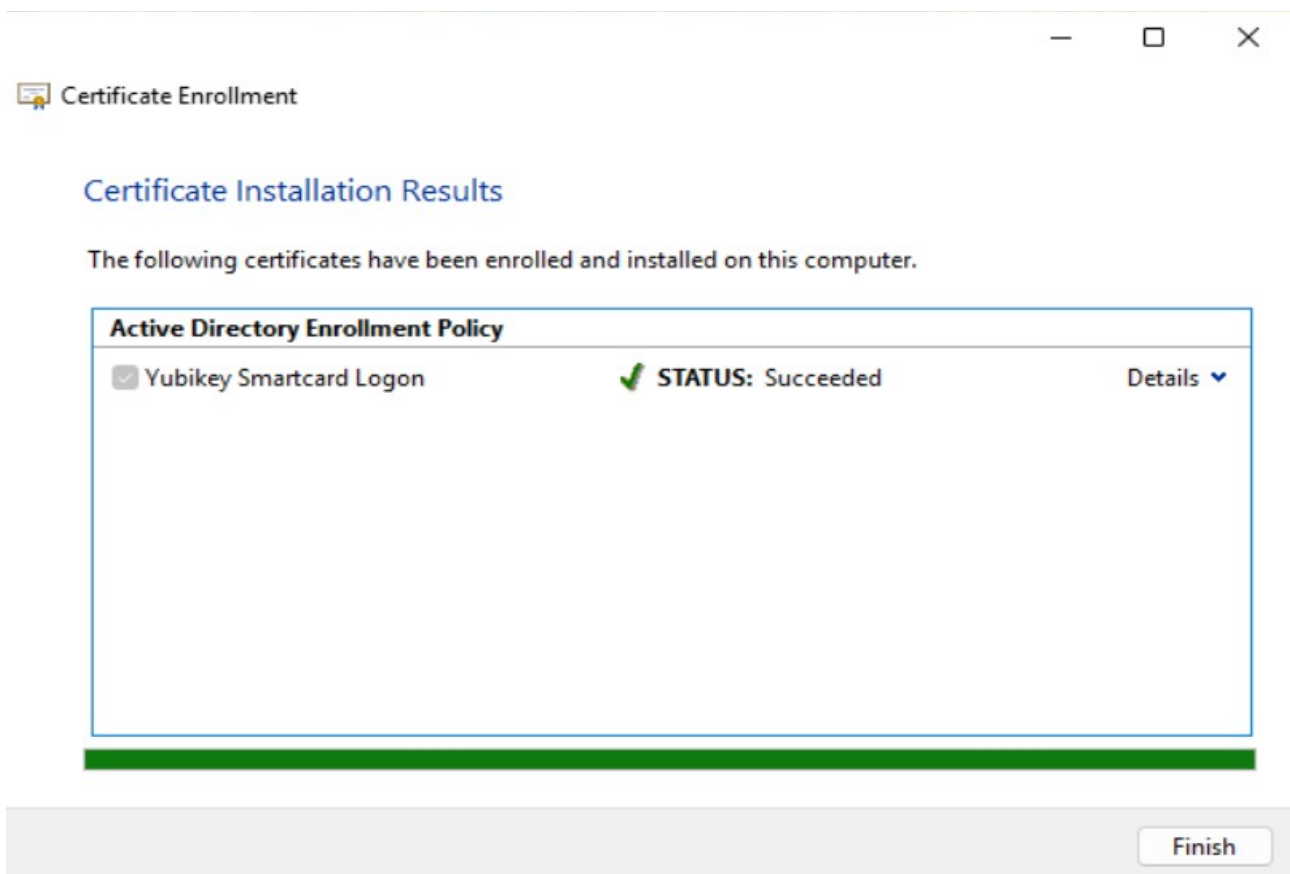
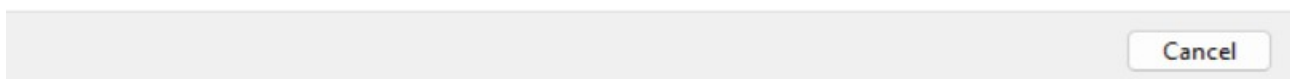
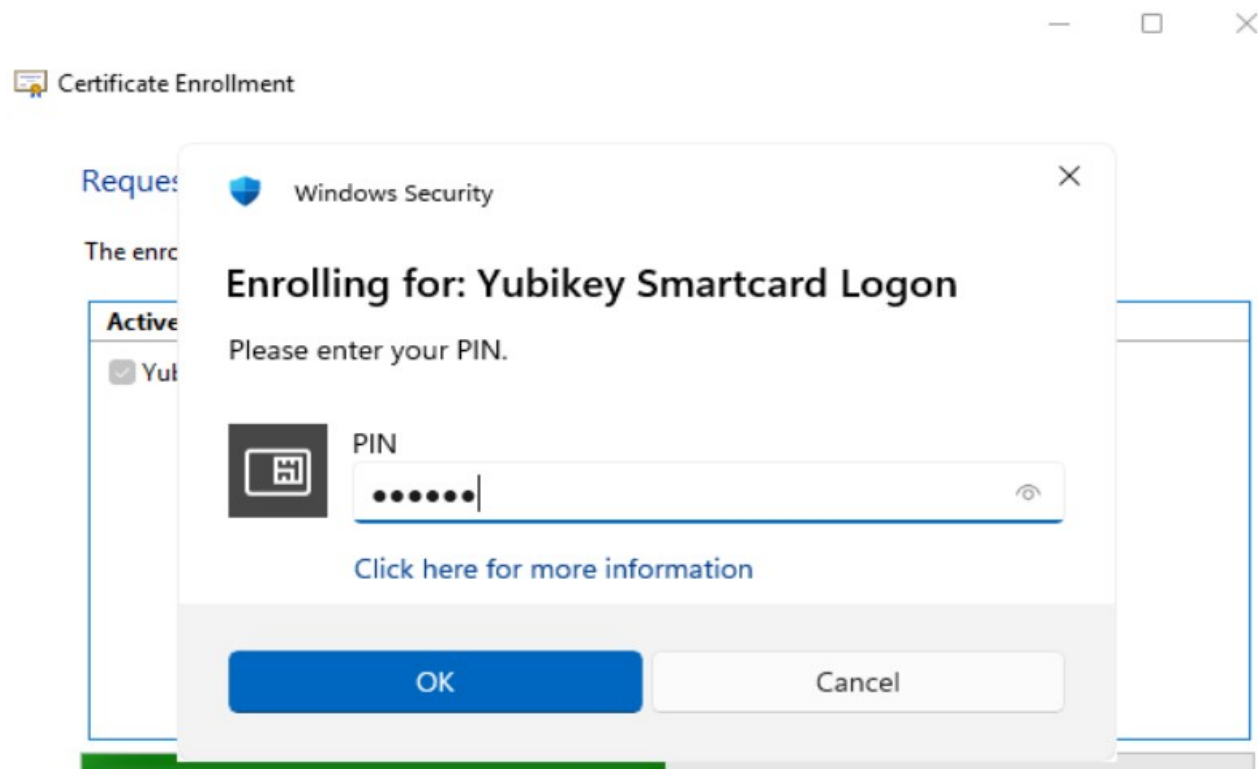
You can request the following types of certificates. Select the certificates you want to request, and then click Enroll.

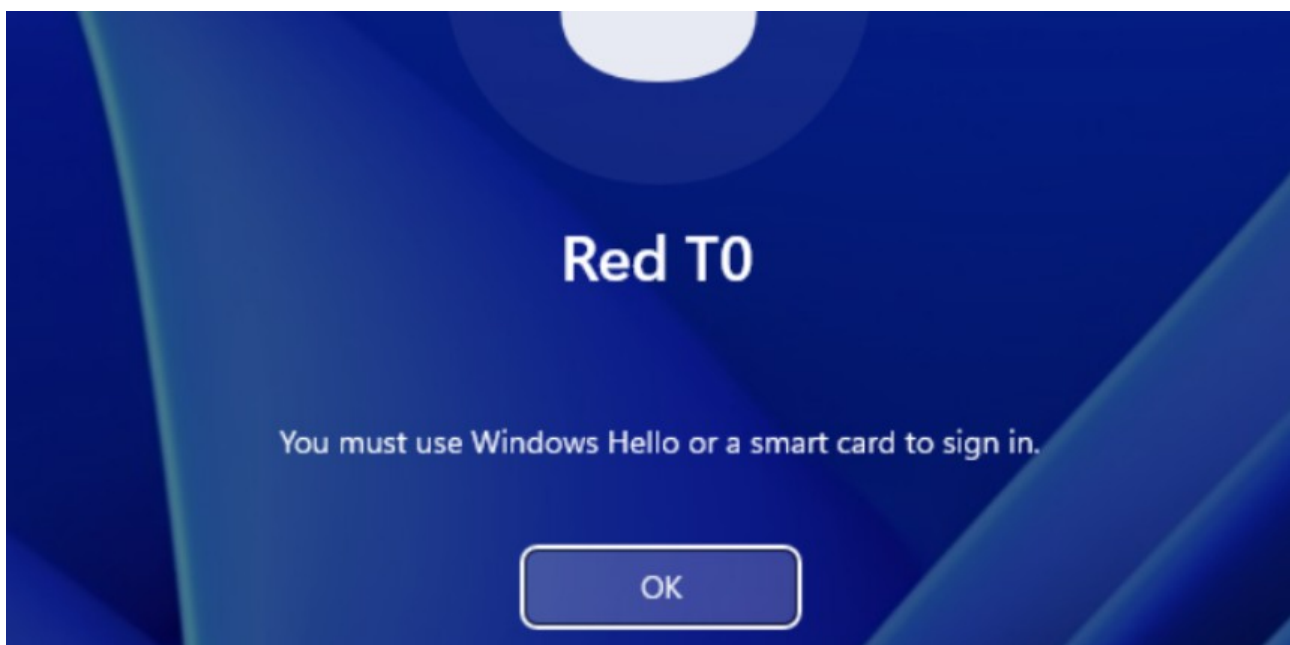
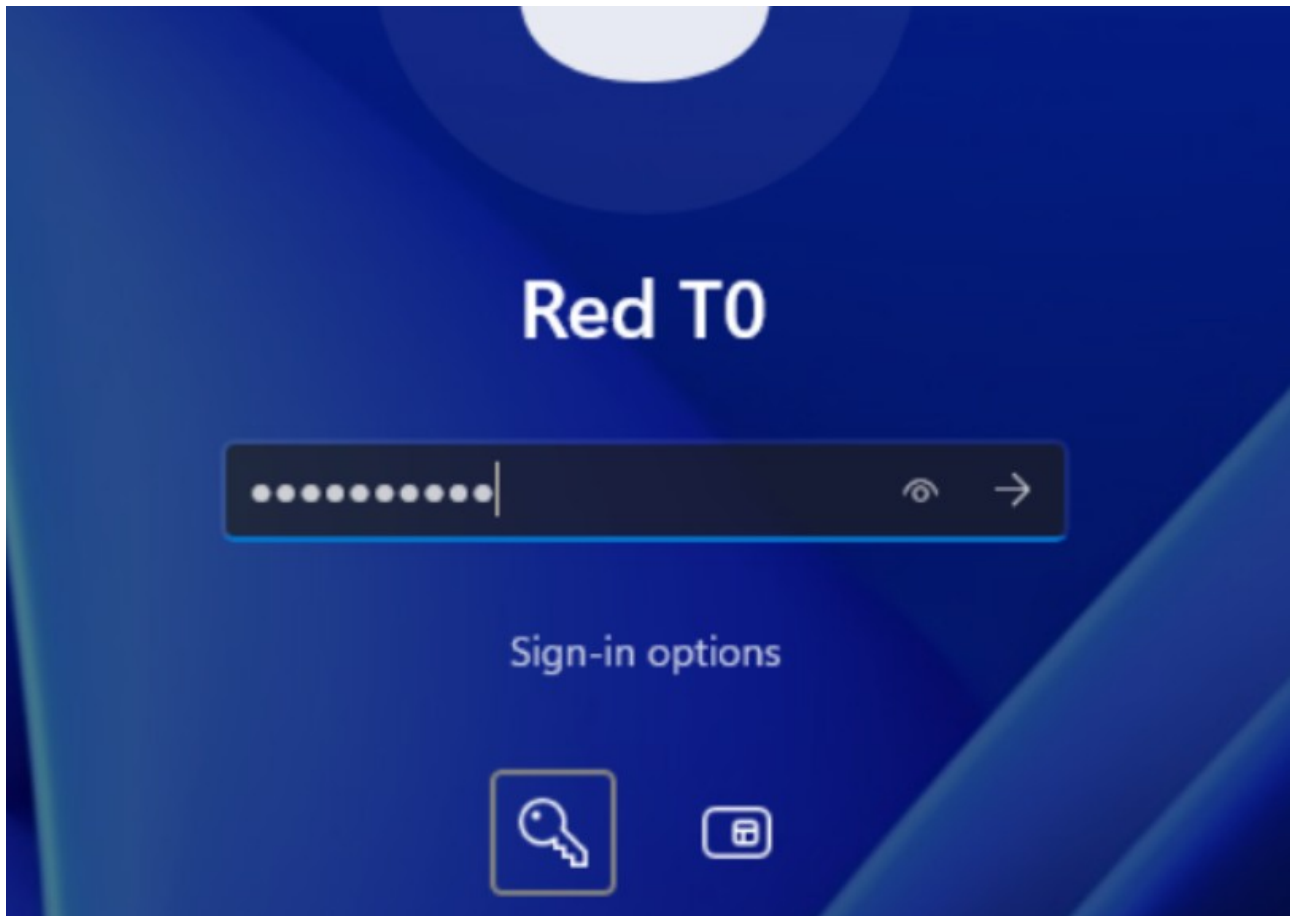
| Active Directory Enrollment Policy                          |                                                                                                              |                                                                                               |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> Basic EFS                          |  <b>STATUS:</b> Available | Details  |
| <input type="checkbox"/> User                               |  <b>STATUS:</b> Available | Details  |
| <input checked="" type="checkbox"/> Yubikey Smartcard Logon |  <b>STATUS:</b> Available | Details  |

☐ Show all templates

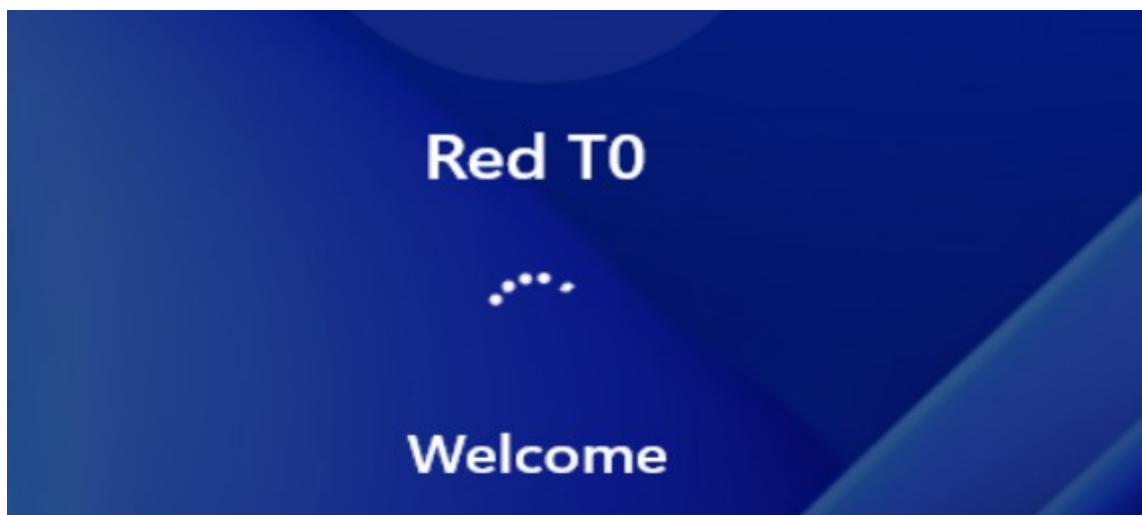
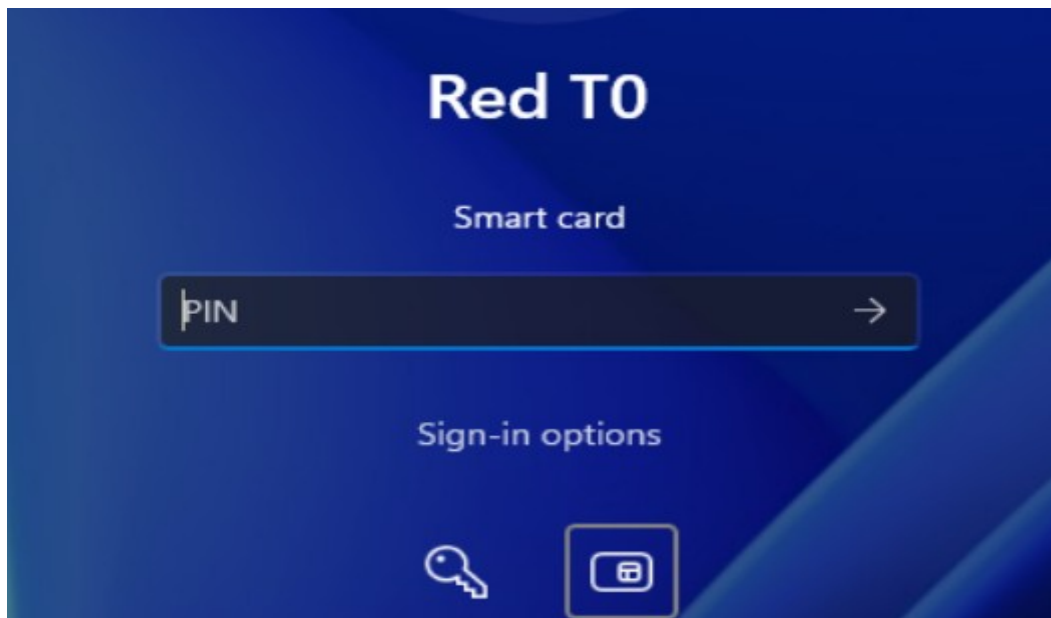
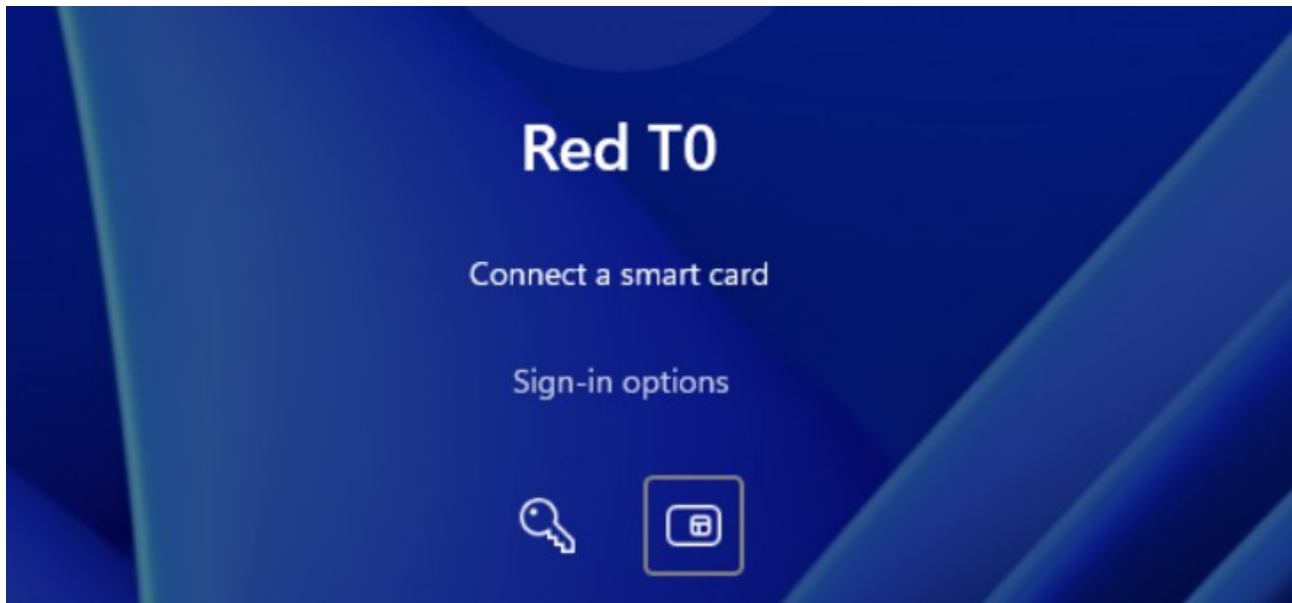
Enroll

Cancel

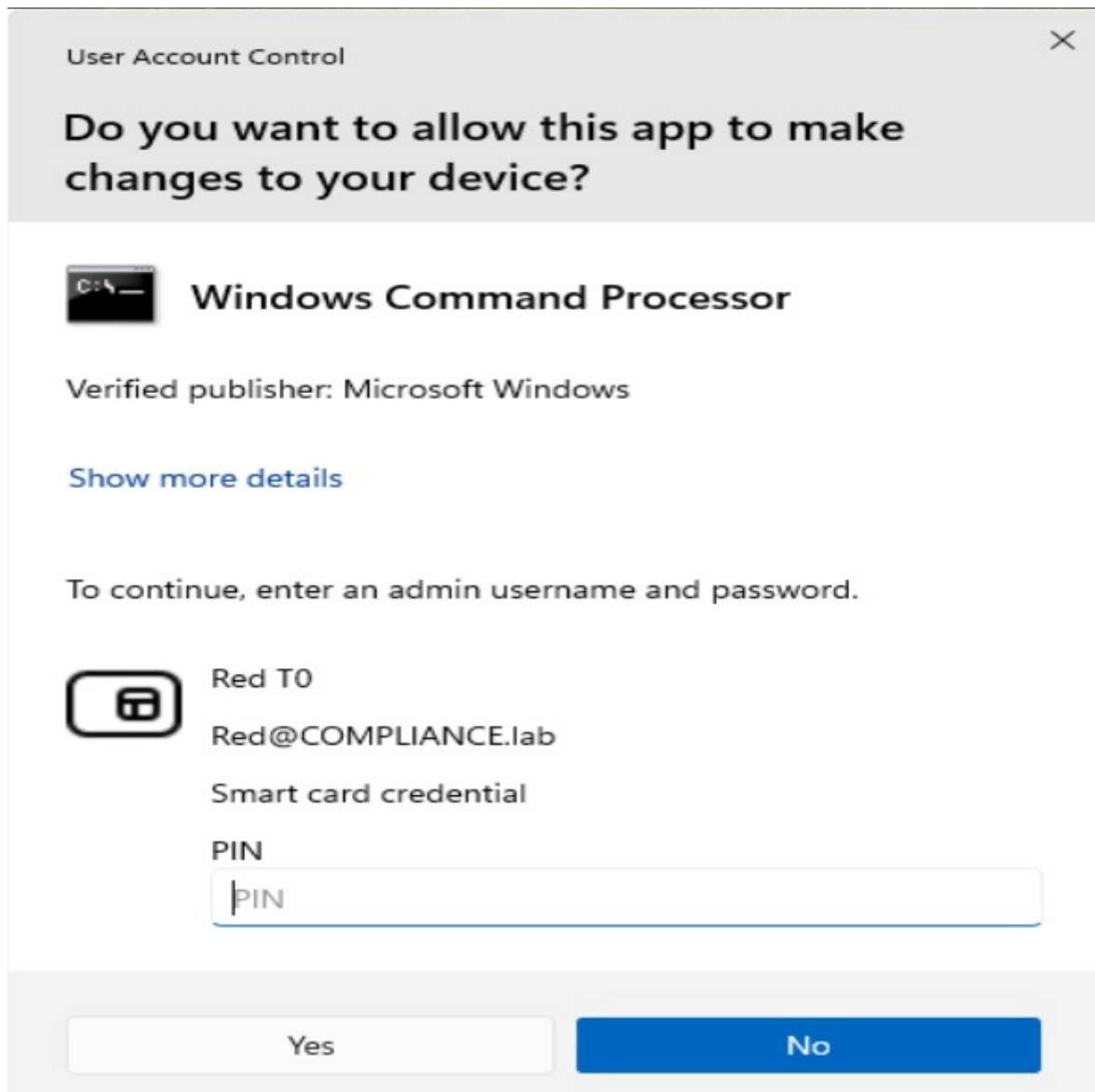






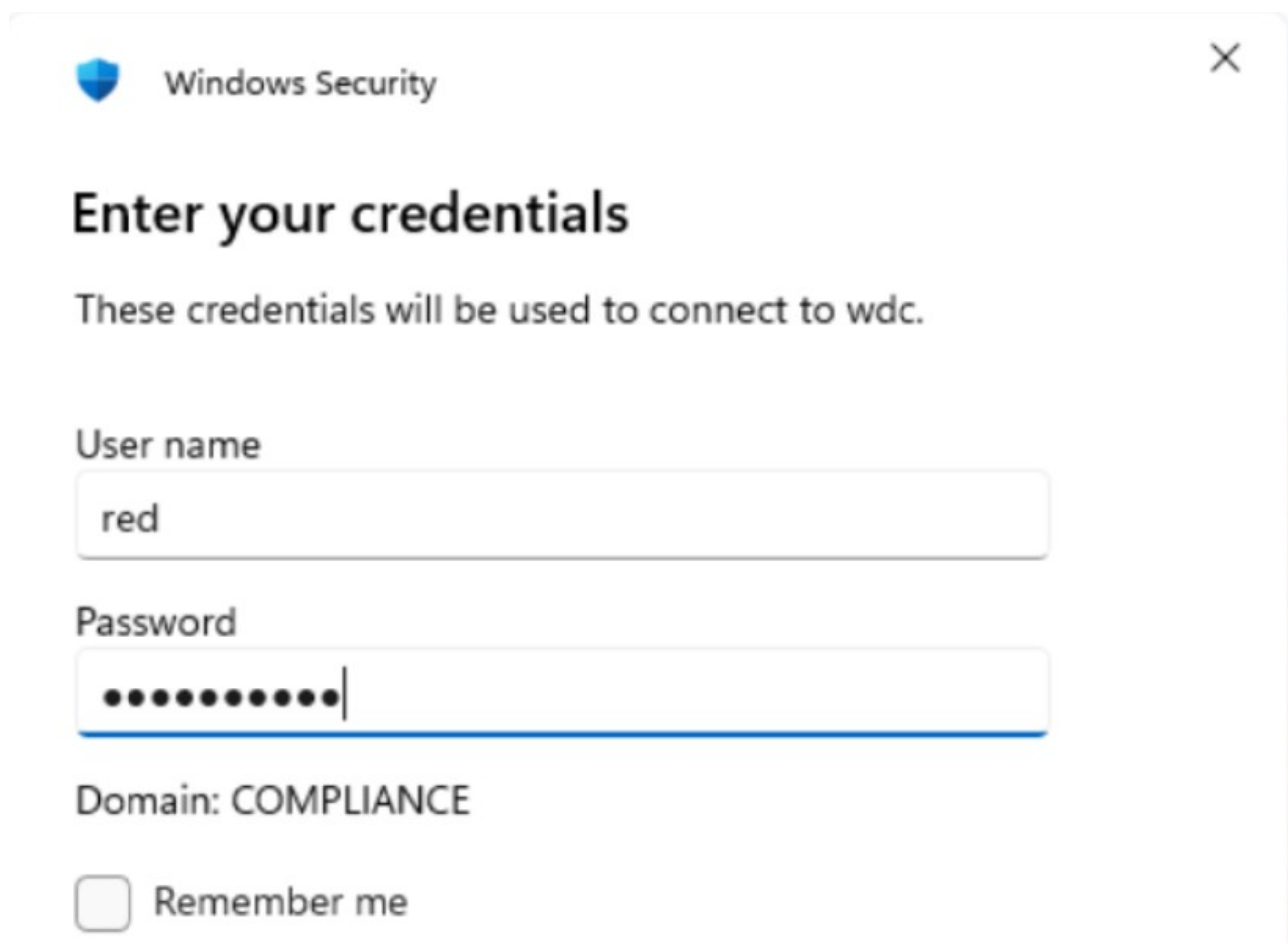


Run CMD as Administrator



Remote Access using RDP





Windows Security

## Enter your credentials

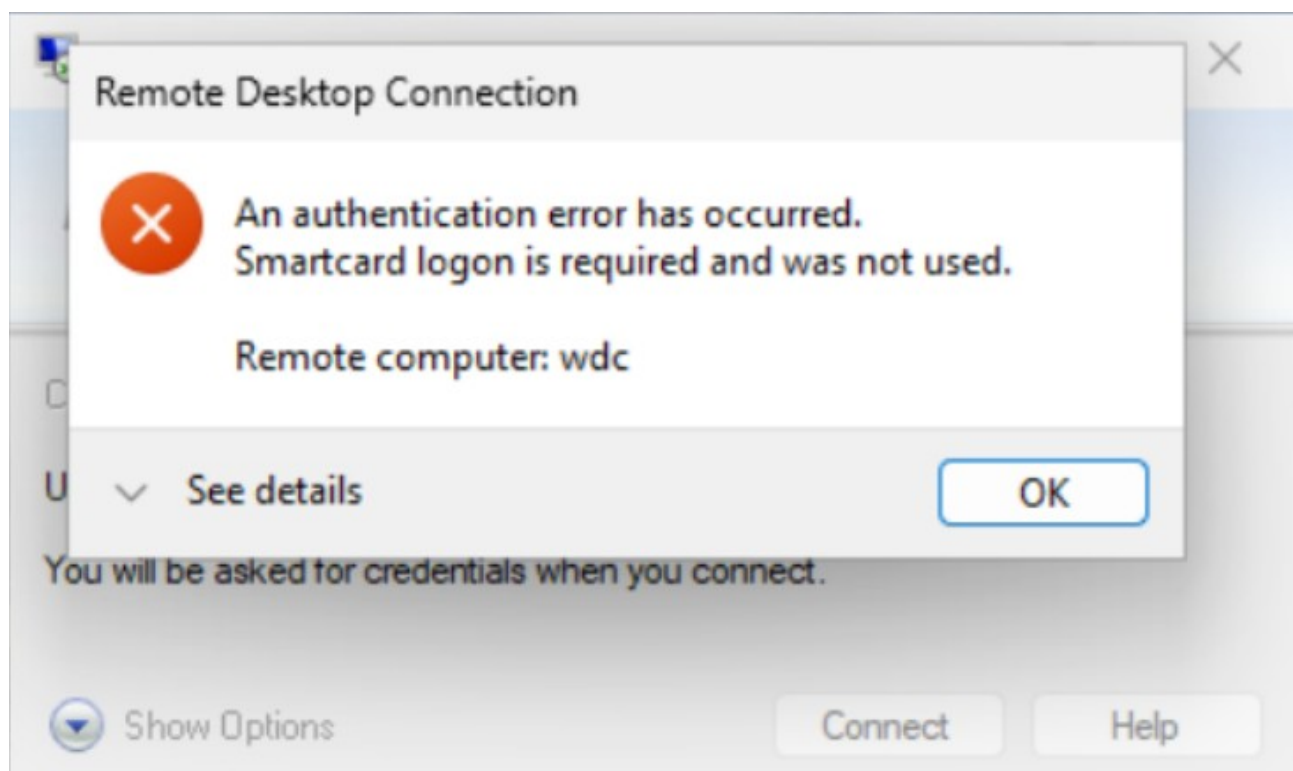
These credentials will be used to connect to wdc.

User name

Password

Domain: COMPLIANCE

☐ Remember me



Remote Desktop Connection

 An authentication error has occurred.  
Smartcard logon is required and was not used.

Remote computer: wdc

U  See details OK

You will be asked for credentials when you connect.

 Show Options Connect Help

 Windows Security ✕

## Enter your credentials

These credentials will be used to connect to wdc.



Red T0

Red@COMPLIANCE.lab

Smart card credential

PIN

••••••

👁

☐ Remember me

[More choices](#)

OK

Cancel

